



The University of Edinburgh



College of Science and Engineering

Mathematics 3 Honours
MATH10003 Financial Mathematics

Friday, 15th December 2017

09:30 – 11:30

Chairman of Examiners – Dr A Olde Daalhuis
External Examiner – Professor J McColl

Answer both questions

Calculators and other electronic aids

A scientific calculator is permitted in this examination.

It must not be a graphical calculator.

It must not be able to communicate with any other devices.

This examination will be marked anonymously.

- (1) (a) Let $(W_t)_{t \geq 0}$ be a standard Brownian motion and c , d and m positive real constants. Calculate

(i) $\mathbb{E}[cW_t + d], \forall t \geq 0$

[1 marks]

(ii) $\mathbb{E}[cW_t^2 + d], \forall t \geq 0$

[2 marks]

(iii) $\mathbb{E}[uW_{\frac{1}{u}} + sW_{\frac{1}{s}}]$, for any $0 < u < s < \infty$

[3 marks]

(iv) $\frac{1}{m^2} \mathbb{E}[W_{m^2s} W_{m^2u}]$, for any $0 < u < s < \infty$

[4 marks]

(v) $\mathbb{E}\left[\left(\frac{1}{m}W_{m^2s} - \frac{1}{m}W_{m^2u}\right)^2\right]$, for any $0 < u < s < \infty$

[4 marks]

- (b) The following SDE with solution $(S_t)_{t \geq 0}$ is the crucial block of the Black-Scholes financial model

$$dS_t = rS_t dt + \sigma S_t dW_t, \quad S_0 = s_0,$$

where $(W_t)_{t \geq 0}$ is a standard Brownian motion, $r, \sigma \in (0, +\infty)$ and $s_0 > 0$ is the initial value.

- (i) Let $(Y_t)_{t \geq 0}$ and $(M_t)_{t \geq 0}$ be two stochastic processes defined as follows:

$$Y_t := e^{-rt} S_t \quad \text{and} \quad M_t := \ln(S_t).$$

Use Itô's formula and derive the SDE that Y_t and M_t satisfy.

[6 marks]

- (ii) For $(M_t)_{t \geq 0}$ defined above, find two deterministic real-valued functions $m : [0, \infty) \rightarrow \mathbb{R}$ and $\gamma : [0, \infty) \rightarrow [0, \infty)$ such that for all $t \geq 0$

$$\frac{M_t - m(t)}{\gamma(t)} \sim \mathcal{N}(0, 1),$$

where $\mathcal{N}(0, 1)$ denotes the standard Normal distribution.

Give short justifications for your calculations.

[5 marks]

- (2) Suppose that $T \in (0, \infty)$, that $(W_t)_{t \in [0, T]}$ is a standard Brownian motion and that the price of a stock at time t is given by

$$S_t = S_0 \exp \left\{ \left(r - \frac{\sigma^2}{2} \right) t + \sigma W_t \right\}, \quad \text{for any } t \in [0, T],$$

for $t \in [0, T]$, where $S_0 > 0$ is the current price of the stock, $r > 0$ is the risk-free interest rate (continuous compounding) and $\sigma \in (0, \infty)$. The bank account pays interest continuously in time at rate r . Assume the market to be free of arbitrage.

Consider a European Call and Put, written on S , maturing at time T and with the same strike $K > 0$. Assume we are working already under the risk-neutral measure.

- (a) (i) State the put-call parity at time $t = 0$ and interpret it. [2 marks]
(ii) State the pricing formula for risk-neutral valuation of a contract of European type. [2 marks]
- (b) In this question one computes the prices of Call and Put options in the Black-Scholes market. With respect to the model of this question, fix the following notation

$$d_+ = \frac{\ln \left(\frac{S_0}{K} \right) + \left(r + \frac{\sigma^2}{2} \right) T}{\sigma \sqrt{T}} \quad \text{and} \quad d_- = \frac{\ln \left(\frac{S_0}{K} \right) + \left(r - \frac{\sigma^2}{2} \right) T}{\sigma \sqrt{T}},$$

where $d_+ = d_- + \sigma \sqrt{T}$ and that $N(x)$ for $x \in \mathbb{R}$ denotes the cdf of the standard Normal distribution.

- (i) Compute the Black-Scholes price of the standard the European Call option $\Phi(S_T) := \max(S_T - K, 0)$ to show that it is given by

$$\Pi(0; \text{Call}(T, K)) = S_0 N(d_+) - K e^{-rT} N(d_-)$$

[8 marks]

- (ii) Compute the Black-Scholes price of the standard the European Put option $\Phi(S_T) := \max(K - S_T, 0)$ to show that it is given by

$$\Pi(0; \text{Put}(T, K)) = K e^{-rT} N(-d_-) - S_0 N(-d_+).$$

[8 marks]

- (iii) Use the explicit expressions for $\Pi(0; \text{Call}(T, K))$ and $\Pi(0; \text{Put}(T, K))$ and show that the Put-Call parity holds in the Black-Scholes market.

[5 marks]